The interpretation algorithm for tbp-parser pipeline output was developed by Microbial Diseases Laboratory Branch (MDL) of California Department of Public Health (CDPH) for integration in the routine TB DST workflow and may not be suitable for all laboratories depending on the availability of confirmatory phenotypic testing and expertise in genetics of TB resistance. The interpretation algorithm is subject to change. For any inquiries, please contact MDL Mycobacterial, Mycotic, and Parasitic Diseases Section, Dr. Varvara Kozyreva (varvara.kozyreva@cdph.ca.gov) and Dr. Matthew Sylvester (matthew.sylvester@cdph.ca.gov)

Introduction

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tbp-parser v3.0.0

WGS- and tNGS-based Drug Susceptibility Testing (DST) assays for *Mycobacterium tuberculosis* complex (MTBC) utilize the same pipeline with some scripts and analysis parameters specific to each of the assays. Mutation interpretation is the same for both WGS-DST and tNGS DST, but quality cutoff might be different.

For both WGS and tNGS, data about the effects of mutations on drug resistance in *Mycobacterium tuberculosis* complex was obtained from WHO, and select peer-reviewed publications based on extensive correlation with phenotypic DST data. If the detected mutation is not found in the database, additional "expert rules" are applied for interpretation of the mutations within genomic regions for which a substantial amount of evidence was accumulated, suggesting likely resistance in case of presence of certain types of mutations anywhere within those regions. As per WHO, the mechanisms by which mutations in such genes confer resistance are well understood, and no epistatic interaction has been observed that could render an isolate with such mutation susceptible. Otherwise, in the absence of strong evidence of association of the detected mutation with either a resistant (R) or susceptible (S) phenotype, mutations are either reported as possessing uncertain significance (U) or not reported at all, depending on the particular gene and level of the confidence in available phenotypic data as per WHO.

In general, if there is an entry in the TBDB database for a mutation, that takes precedence and is applied first. Validated TBDB database includes only mutations included in WHO catalogue of mutations v2 (2023).

The interpretation logic and expert rules are implemented in the tbp-parser tool available on GitHub and are described below.

For more information on CDPH MDL laboratory processes, input parameters, and additional quality reporting requirements that are not implemented in tbp-parser, please see the complementary Analysis Principles PDF file for the corresponding tbp-parser version.

WGS- and tNGS-DST interpretation summary

	Individual mutation reporting		Drug interpretation reporting
Mutation Interpretation (in order of highest to lowest severity)	**Format of individual mutations reported per gene locus[1]**	**Mutation is listed in clinical report (Yes/No)**	**Overall value based on the highest severity mutation within all targets associated with given drug**
R **Mutations for all targets, except for *rpoB* (see below)**	p.Ala689Val (AA) or c.-40C>T (nt)	Yes	Mutation(s) associated with resistance to XXX detected
R **Mutations in *rpoB* gene: Low-level RIF R mutations**	p.Ala689Val	Yes	Predicted low-level resistance to rifampin. May test susceptible by phenotypic methods
R **Mutations in *rpoB* gene: Other RIF R mutations**	p.Ala689Val	Yes	Predicted resistance to rifampin
Insufficient Coverage in the gene locus[2] **AND** successfully sequenced areas of the same gene **OR** other genes associated with given drug do **NOT** contain mutations interpreted as R	No sequence[3]	N/A	Pending Retest
U	p.Ala689Val (AA) or c.-40C>T (nt)	Yes	The detected mutation(s) have uncertain significance. Resistance to XXX cannot be ruled out
S **Mutations for all targets, except for *rpoB* (see below)**	No high confidence mutations detected	No	No mutations associated with resistance to XXX detected
S **Mutations in *rpoB* gene: Synonymous mutation present within *rpoB* RRDR (codons 426-452)**	p.Ala689Ala [synonymous][4]	Yes	Predicted susceptibility to rifampin. The detected synonymous mutation(s) do not confer resistance[5]
S **Mutations in *rpoB* gene: Other RIF S mutations (outside of RRDR)**	No high confidence mutations detected	No	Predicted susceptibility to rifampin[6]
WT	No mutations detected	N/A	No mutations associated with resistance to XXX detected

[1] Amino acid (AA) mutation format is displayed preferentially; if it is not available (e.g., noncoding gene regions and promoters), mutations are displayed in nucleotide (nt) format.
[2] Deletions and areas of poor sequencing coverage are differentiated by tbp-parser.
[3] If the gene has poor locus coverage but has "R" mutation(s) that pass positional QC thresholds within the same gene, report those mutations and corresponding drug interpretation based on the "R" mutation(s).
[4] Displayed in addition to any "U" or "R" mutations detected in RRDR.
[5] If other *rpoB* mutations have interpretation "U" or "R", then report drug predicted susceptibility based on those mutations' interpretation.

Quality Control Input Parameters

The following input parameters affect **quality control**:

- **min_depth** (default 10x): the minimum depth of coverage for a site to pass QC
- **min_percent_coverage** (default 100%): the minimum percentage of a region that has depth **above** the threshold set by min_depth to pass QC
- **min_read_support** (default 10): the minimum read support (calculated by multiplying a mutation's depth by its frequency) for a mutation to pass QC
- **min_frequency** (default 10%): the minimum alternate allele frequency for a mutation to pass QC
- **min_percent_loci_covered** (default 70%): the minimum percentage of the genes/loci included in the LIMS report that must pass coverage QC for the sample to be identified as MTBC

The following input parameters impact **the regions used for quality control**:

- **coverage_bed** (default is provided; see GitHub): the BED file containing the genes of interest, their locus tags, and their regions for QC calculations.
- **err_coverage_bed** (no default): an optional BED file containing narrower ranges that are essential for resistance reporting instead of the entire gene locus; for advanced usage. ***If provided, deletions that are flagged in the Coverage report were valid in the ERR***.
- **use_err_for_qc** (default false): an option to instruct tbp-parser to use the ERR regions instead of the regions specified by the coverage_bed file. The regions indicated by the coverage_bed file will still be reported in the coverage report, but all QC for the laboratorian report and LIMS report will be conducted on the data for the ERR regions.

The following input parameters are **specific for tNGS**; they are **not** compatible with WGS data.

- **tngs** (default false): an option to instruct tbp-parser to use tNGS-specific logic (lineage designation and QC); sections in the interpretation logic that say "tNGS-specific" are only implemented if this parameter is set to true.
- **tngs_read_support_boundaries** (default 10, 10): the upper and lower read support boundaries for tNGS QC reporting.
- **tngs_frequency_boundaries** (default 10%, 10%): the upper and lower frequency boundaries for tNGS QC reporting.

The boundary parameters are unique in that they are used together. A mutation will pass QC if either of the following conditions is true:

Conditions for QC pass	
Condition 1	Lower Boundary ≤ Read Support ≤ Upper Boundary
	Frequency ≥ Upper Boundary
Condition 2	Read Support ≥ Upper Boundary
	Frequency ≥ Lower Boundary

The default boundary parameters are set so that behavior is the same as the standard QC checks. Modification of these boundaries will alter QC determinations from the standard.

Interpretation Logic

In general, if there is an entry in the TBDB database (based on the WHO v2 catalogue) for a mutation, that takes precedence and is applied first. TBDB database includes all mutations in the `mutations.csv` file.

Sections 1-4 provide detailed descriptions of the parsing that is performed to generate interpretations for detected mutations, also referred to as the "Laboratorian report" and contains broader information than is reported as a part of clinical report (also known as the LIMS report; see section 5).

Formatting keys:
- Items written in `Consolas` refer to fields extracted from TBProfiler output JSONs.
- Items written in blue refer to output fields in the generated reports
- Items written in orange refer to values determined by input parameters

1. For **genes** ***mmpR5 (Rv0678), atpE, pepQ, mmpL5, mmpS5, rrl, rplC*** ***(genes related to new drugs based on CDC expert rules)****:*

   1.1. IF mutation is in either of *mmpR5 (Rv0678), atpE, pepQ, mmpL5, mmpS5, rrl, rplC*
   AND `confidence` has value for the corresponding drug,
   THEN copy that value to the "confidence" column
   AND assign the following "looker_interpretation" & "mdl_interpretation" based on that confidence value:

Mutation classification (`confidence` field)	**Looker interpretation**	**MDL Interpretation**
WHO: associated with R [6]	R	R
WHO: associated with R- interim [7]	R-Interim	R
WHO: uncertain [8,9]	U	U
WHO: NOT associated w/R [10]	S	S
WHO: NOT associated w/R- interim [11]	S-Interim	S
WHO: synonymous [12]	S	S

   1.2. ELSE IF mutation is in either of *mmpR5 (Rv0678), atpE, pepQ, mmpL5, mmpS5, rrl, rplC*
   AND NO `confidence` available for the corresponding drug
   THEN assign confidence = "No WHO annotation"
   AND assign the "looker_interpretation" & "mdl_interpretation" value based on the table below:

[6] Exact value in WHO & TBDB "Assoc w R"
[7] Exact value in WHO & TBDB "Assoc w R – Interim" or "Assoc w R – interim"
[8] Exact value in WHO & TBDB "Uncertain significance"
[9] WHO also has value "NA" that is not found in TBDB
[10] Exact value in WHO "Not assoc w R"; not in TBDB
[11] Exact value in WHO & TBDB "Not assoc w R – Interim" or "Not assoc w R – interim"
[12] Exact value in WHO "Synonymous"; not in TBDB

Genes	Mutation location	Mutation type	Looker interpretation	MDL Interpretation
mmpR5 (Rv0678), atpE, pepQ, rplC, mmpS5	Target Promoter[13]	any	U	U
	Non-target promoter[14]	any	S	S
	ORF (everything else)	NONsyn	U	U
		syn	S	S
mmpL5	Non-target promoter[15]	any	S	S
	ORF (everything else)	NONsyn	U	U
		syn	S	S
rrl	Target Promoter[16]	any	U	U
	nt positions 2003-2367 and 2449-3056 in rRNA	any	U	U
	outside of nt positions 2003-2367 and 2449-3056 in rRNA	any	S	S

1.3. Only for genes *mmpR5 (Rv0678), mmpL5, mmpS5* – to standardize the output and interpretation of mutations that may be listed in relation to any of those 3 genes:

Output all mutations in "`consequences`" field to get information about these mutations in relation to the other three genes
AND follow rule 1.2 to assign "looker_interpretation" & "mdl_interpretation"

2. For **genes *katG, pncA, ethA, gid, rpoB (covered by WHO expert rules):***

2.1. IF mutation is in either ***katG***, ***pncA***, ***ethA***, ***gid***, or ***rpoB***,
AND `confidence` has value for the corresponding drug,
THEN copy that value to the"confidence" column
AND assign the following "looker_interpretation" & "mdl_interpretation" based on that confidence value:

Mutation classification (`confidence` field)	**Looker interpretation**	**MDL Interpretation**
WHO: associated with R	R	R
WHO: associated with R- interim	R-Interim	R
WHO: uncertain	U	U
WHO: NOT associated w/R	S	S
WHO: NOT associated w/R- interim	S-Interim	S
WHO: synonymous	S	S

[13] See coordinates in Table 1 at end of document
[14] Mutation effect `type` is "upstream_gene_variant", but does not fall within the target promoter region
[15] Mutation effect `type` is "upstream_gene_variant", but does not fall within the target promoter region
[16] See coordinates in Table 1 at end of document

2.2. ELSE IF mutation found in either ***katG***, ***pncA***, ***ethA***, ***gid***, or ***rpoB***,
AND no `confidence` available for the corresponding drug,
THEN apply WHO Expert rule:

2.2.1. IF mutation found in ***katG***, ***pncA***, ***ethA***, or ***gid*** (not *rpoB*):
AND mutation represents a loss-of-function (mutation contains "del", "ins", "fs", "delins", "_" or ends with "*")
AND those mutations are found in ORF or within first 30 nucleotides upstream start codon[17]
THEN assign confidence = "No WHO annotation"
AND assign the "looker_interpretation" & "mdl_interpretation" = "**R**" for corresponding mutation.
ELSE assign confidence = "No WHO annotation"
AND assign the "looker_interpretation" & "mdl_interpretation" value based on the table below:

Mutation classification	Looker interpretation	MDL Interpretation
Synonymous	S	S
NONsynonymous[18]	U	U
Mutation in Target Promoter[19]	U	U
Non-target promoter[20]	S	S

2.2.2. ELSE IF mutation found in ***rpoB***:

2.2.2.1. **IF** mutation within *rpoB* codons 426-452 (even partially, if it's an indel)
THEN assign confidence = "No WHO annotation"
AND assign the "looker_interpretation" & "mdl_interpretation" value based on table below:

Mutation classification	Looker interpretation	MDL Interpretation
Synonymous	S	S
NONsynonymous[21]	R	R

2.2.2.2. **ELSE IF** mutation outside of codons 426-452
THEN assign confidence = "No WHO annotation"
AND assign the "looker_interpretation" & "mdl_interpretation" value based on the table below:

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[17] Logic 2.2.1.1. overrules any Table 1 proximal promoter regions logic. The expected behavior:
katG mutation that is not LoF between -1 to -30 bp = U
katG mutation that is not LoF between -30 to -532 bp = U
katG mutation that is not LoF further than -532 bp = S
katG mutation that is LoF between -1 to -30 bp = R
katG mutation that is LoF between -30 to -532 bp = U
katG mutation that is LoF further than -532 bp = S

[18] Including indels in ORF

[19] See coordinates in Table 1 at end of document

[20] Mutation effect `type` is "upstream_gene_variant", but does not fall within the target promoter region

[21] Including indels in ORF

Mutation classification	Looker interpretation	MDL Interpretation
Synonymous	S	S
NONsynonymous[22]	U	U
Mutation in Target Promoter[23]	U	U
Non-target promoter[24]	S	S

3. For genes **other than** *mmpR5 (Rv0678), atpE, pepQ, mmpL5, mmpS5, rrl, rplC, katG, pncA, ethA, gid, rpoB*:

3.1. IF mutation is in gene other than *mmpR5 (Rv0678), atpE, pepQ, mmpL5, mmpS5, rrl, rplC, katG, pncA, ethA, gid, rpoB*
AND `confidence` has value for the corresponding drug,
THEN keep that value as in "confidence" column
AND assign the following "looker_interpretation" & "mdl_interpretation" based on that confidence value:

Mutation classification (`confidence` field)	Looker interpretation	MDL Interpretation
WHO: associated with R	R	R
WHO: associated with R- interim	R-Interim	R
WHO: uncertain	U	U
WHO: NOT associated w/R	S	S
WHO: NOT associated w/R- interim	S-Interim	S
WHO: synonymous	S	S

3.2. ELSE IF mutation is in gene other than *mmpR5 (Rv0678), atpE, pepQ, mmpL5, mmpS5, rrl, rplC, katG, pncA, ethA, gid, rpoB*
AND NO `confidence` available for the corresponding drug:

3.2.1. IF Mutation is located within ***rrs*** gene region,

3.2.1.1. **IF** Mutations at *rrs* nucleotide positions 1401, 1402, or 1484 (positions in relation to NC_000962.3: 1473246, 1473247, or 1473329)
THEN assign:
confidence = "No WHO annotation"
looker_interpretation = U
mdl_interpretation = U

3.2.1.2. **ELSE** assign:
confidence = "No WHO annotation"
looker_interpretation = S
mdl_interpretation = S

[22] Including indels in ORF
[23] See coordinates in Table 1 at end of document
[24] Mutation effect `type` is "upstream_gene_variant", but does not fall within the target promoter region

3.2.2. ELSE IF Mutation is located within ***gyrA*** coding gene (GyrA QRDR expert rule):
AND mutation is within region codons 88-94 of *gyrA*
AND mutation is nonsynonymous
THEN assign:
confidence = "No WHO annotation"
looker_interpretation = U
mdl_interpretation = U

3.2.3. ELSE IF Mutation is located within ***gyrB*** coding gene (GyrB QRDR expert rule):
AND mutation is within region codons 446-507 of *gyrB*
AND mutation is nonsynonymous
THEN assign:
confidence = "No WHO annotation"
looker_interpretation = U
mdl_interpretation = U

3.2.4. ELSE (all remaining scenarios not covered above),
THEN assign confidence = "No WHO annotation"
AND assign the "looker_interpretation" & "mdl_interpretation" value based on table below:

Mutation classification	Looker interpretation	MDL Interpretation
Synonymous	S	S
NONsynonymous[25]	U	U
Mutation in Target Promoter[26]	U	U
Non-target promoter[27]	S	S

4. Reporting of remaining scenarios:

4.1. IF No mutations detected in given gene (**WILD TYPE**)
AND gene locus passes locus QC (see 4.2.2)
THEN add that gene at the bottom of laboratorian report
AND assign:
tbprofiler_variant_substitution_nt = "WT"
tbprofiler_variant_substitution_aa = "WT"
tbprofiler_variant_substitution_type = "WT"
tbprofiler_variant_position = "NA"
confidence, depth, frequency, read_support, rationale = "NA"
antimicrobial = corresponding drug to which this gene locus confers resistance [28]
looker_interpretation = "S"
mdl_interpretation = "WT"

[25] Including indels in ORF
[26] See coordinates in Table 1 at end of document
[27] Mutation effect `type` is "upstream_gene_variant", but does not fall within the target promoter region
[28] The same locus may be listed more than once as "WT" for each of the drugs that it is associated with, since mutations in some loci cause R to more than one drug

4.2. Quality filtering of final reportable results:

4.2.1. Positional QC (regarding the position where mutation is detected):

Conditions for QC pass
≥ min_depth
≥ min_frequency
≥ min_read_support

Additional tNGS-specific QC using the tNGS boundary arguments:

Conditions for QC pass	
Condition 1	Lower Boundary ≤ Read Support < Upper Boundary
	Frequency ≥ Upper Boundary
Condition 2	Read Support ≥ Upper Boundary
	Frequency ≥ Lower Boundary

4.2.1.1. **IF** Detected mutation that is **NOT** a Deletion **FAILS** positional QC **THEN** add "Failed quality in the mutation position" in the warning field for the corresponding mutation.

4.2.1.2. **ELSE IF** Detected **DELETION** has Variant Read Depth > 0 **AND** < min_depth for WGS **THEN** write "Failed quality in the mutation position" in the warning field for the corresponding mutation.

4.2.1.3. **ELSE IF** Detected **DELETION** has Variant Read Depth = 0 **AND** passes min_frequency [29] **THEN** report mutation on LIMS report

4.2.1.4. **ELSE** report mutation on LIMS report

4.2.2. Locus QC (regarding the breadth of coverage throughout the locus):

Locus QC	Conditions	Note
PASS	≥ min_percent_coverage	
	< min_percent_coverage **AND** a valid deletion is **present**	Valid deletions pass positional QC (see 4.2.1) **AND** are at least partially in the ORF; the "Insufficient coverage for locus" warning is still added
FAIL	< min_percent_coverage **AND** a valid deletion is **absent**	

IF locus QC check **FAILS**:

[29] This condition exists because TBProfiler v6.6.3 does not correctly evaluate depth of coverage for these deletions

4.2.2.1. **IF** No mutations detected (WT)
THEN add "Insufficient coverage for the locus" to the warning field for the corresponding gene
AND *overwrite* the following values for that gene (found at the bottom of laboratorian report):
looker_interpretation = "Insufficient Coverage"
mdl_interpretation = "Insufficient Coverage"
Do not change any other fields from what they normally would be for a WT locus.

4.2.2.2. **ELSE IF** only "S" or "U" mutations are detected in given gene (i.e., no "R" mutations detected for the gene in the "mdl_interpretation" field)
THEN add "Insufficient coverage for the locus" to the warning field for the corresponding detected S or U mutation, if any [30]
AND *overwrite* the following values:
looker_interpretation = "Insufficient Coverage"
mdl_interpretation = "Insufficient Coverage"
Do not change any other fields from what they normally would be for the detected mutation.

4.2.2.3. **ELSE IF** R mutation is detected in given gene (based on mutation interpretation in "mdl_interpretation" field)
AND **NO** "Failed quality in the mutation position" in the warning field
THEN add "Insufficient coverage for the locus" to the warning field
Do not overwrite Looker/MDL interpretation columns.

4.2.2.4. **ELSE IF** R mutation detected in given gene (based on mutation interpretation in "mdl_interpretation" field)
AND "Failed quality in the mutation position" in the warning field
THEN add "Insufficient coverage for the locus" to the warning field for the corresponding detected mutation [31]
AND *overwrite* the following values:
looker_interpretation = "Insufficient Coverage"
mdl_interpretation = "Insufficient Coverage"
Do not change any other fields from what they normally would be for the detected mutation.

4.3. Additional analysis notes:

4.3.1. Indels that start outside of target regions that are a subject of an expert rule, but span into the region of interest, are interpreted as meeting the corresponding rule. This includes indels that span the entire region of interest.

For example, a deletion in rplC that occurs over -60 to -48 partially falls into the target promoter region coordinates for that gene (-51 to -1). This deletion is then treated with the target promoter region expert rule (if no WHO confidence available).

[30] Do not overwrite positional QC warnings for the variant; add further warnings *in addition*
[31] Do not overwrite positional QC warnings for the variant; add further warnings *in addition*

4.3.2. All antimicrobial drugs associated with a gene are reported. This includes WHO annotation and the `gene_associated_drugs` field.

If there is a mutation in the gene associated with two drugs (e.g. Rv0678 for BDQ and CFZ) and TBDB only has confidence listed for one drug and not another drug, the following logic is implemented:

a. Generate interpretation based on `confidence` in TBDB for the drug in TBDB;
b. Follow the applicable expert rule for the drug that is not listed in TBDB.

4.3.3. If multiple identical variants exist (same gene, position, and drug), keep the variant with the highest-severity WHO confidence ranking (from high to low: **Assoc w R > Uncertain Significance > Not Assoc w R > No WHO annotation**).

5. LIMS report parsing:

5.1. Below is description of how a separate output file is generated for ingest into LIMS[32] by parsing results of described above interpretation output (aka "Laboratorian report") for resistance reporting (5.2 – 5.3), and parsing TBProfiler output for species ID (5.4).

NOTE: Mutations that failed positional QC (even if locus QC is passing) **DO NOT** appear on LIMS report and **DO NOT** affect drug interpretation[33].

5.2. Outputting ***drug interpretations*** in LIMS report (e.g. INH): an overall value based on the highest-severity mutation (from high to low: **R > Insufficient Coverage > U > S > WT)** is generated. Assign text value for corresponding drug according to the table below:

Information coming from Laboratorian report			**In LIMS output of TB Profiler**
MDL Interpretation (overall value based on the highest severity mutation within all targets associated with given drug)			**Drug (e.g. INH)**
R	R for all drugs, except RIF (see below)		Mutation(s) associated with resistance to XXX detected
	R mutations in *rpoB* gene	Following mutations in *rpoB* gene: Leu430Pro Asp435Tyr His445Asn His445Ser His445Leu His445Cys Leu452Pro Ile491Phe **AND** no other R mutations in *rpoB* gene	Predicted low-level resistance to rifampin. May test susceptible by phenotypic methods
		All other R mutations in *rpoB* gene	Predicted resistance to rifampin

[32] LIMS output was originally designed for CalLIMS

[33] For example, if a mutation is S and failed positional QC, it will **NOT** affect LIMS drug interpretation. The gene result will display "No mutations" (if no other higher severity mutations are found in the gene).

"Insufficient Coverage" for **any** gene(s) associated with given drug **AND** no other genes associated with given drug has mutation interpreted as R			Pending Retest (If other genes associated with the same drug have mutations with MDL interpretation "R", then report drug predicted susceptibility based on those targets)
U			The detected mutation(s) have uncertain significance. Resistance to XXX cannot be ruled out
S	S for all drugs, except RIF (see below)		No mutations associated with resistance to XXX detected
	S interpretation for RIF	Synonymous mutation present within *rpoB* codons 426-452 (RRDR)	Predicted susceptibility to rifampin. The detected synonymous mutation(s) do not confer resistance
		All other cases for S mutations in *rpoB*	Predicted susceptibility to rifampin
WT (no mutations detected in the corresponding targets)			No mutations associated with resistance to XXX detected

5.2.1. Drug interpretation is based only on the set of genes that is being reported. So for LIMS, if a gene is not on the reportable list, it is ignored. Overall drug interpretation is generated based on mutations found (or not) in the genes listed in LIMS report.

5.2.2. Column "Lineage" in LIMS report is populated from the `main_lineage` output of TB Profiler. This information is not clinically reportable.

5.3. Outputting ***individual mutations*** in gene target fields in LIMS report (e.g. INH_katG):

5.3.1. Only output mutations classified as "R" or "U" in "mdl_interpretation" field of Laboratorian report, for the exception of RRDR region of *rpoB*:

5.3.1.1. Within *rpoB* codons 426-452, output ALL individual mutations (including synonymous). Add "[synonymous]" after any synonymous mutation output text; e.g., "p.Asp877Asp [synonymous]"

(RIF interpretation output is changed accordingly in the case if the only mutation detected in rpoB RRDR is synonymous; see above)

5.3.2. See instructions on output format for individual mutations below. Report the amino acid change preferentially; only use the nucleotide change if an amino acid change does not exist (in promoters, etc.)

Information coming from Laboratorian report	**In LIMS output of tbp-parser**

MDL Interpretation (for each given mutation)		**Gene Loci (e.g. INH_katG), i.e. where individual mutations that are listed for each target**	
		Format	**Mutation should be listed in LIMS report**
R		Report the detected mutation in format " p.Ala689Val" [34]	Yes
"Insufficient Coverage"		Type "No sequence" for the corresponding gene target (unless R mutation(s) with adequate coverage were detected in gene)	N/A
U		Report the detected mutation in format "p.Ala689Val"	Yes
S	S for mutations in RRDR region of *rpoB* gene: If synonymous mutation is present within *rpoB* codons 426-452 (Do not report synonymous mutations outside this region)	Report the detected synonymous mutation in format " p.Ala689Ala [synonymous]"	Yes
	S – for all other cases	Mutation is not displayed on LIMS report. If the only mutation(s) detected in given target gene were S, type "No high confidence mutations detected"	No
WT (no valid[35] mutations detected in the corresponding target)		Type "No mutations detected" for the corresponding gene target	N/A

5.3.3. If a mutation in the same position (usually a deletion) is repeated but with different annotation types on Laboratorian report, use the one with highest read_support to decide which deletion to report in LIMS. Leave the alternative mutation annotation on Laboratorian report but do not use it for evaluation of predicted drug resistance.

5.4. **<u>MTBC ID by WGS only</u>**: Parse the TBProfiler "`main_lineage`" field original output and the Coverage report to populate the LIMS report field "Lineage ID" for species ID. The value reported depends on the text identified in the "`main_lineage`" field.

5.4.1. IF ≥ min_percent_loci_covered of genes on the Coverage report that are ALSO on the LIMS report have ≥ min_percent_coverage (per percent_coverage field), **THEN** assign Lineage ID based on the following table:

[34] If multiple mutations in the gene, separate them by semicolons: "p.Ala689Val; p.His57Asp"
[35] A valid mutation is a mutation that passes QC

Information coming from TBProfiler	Lineage ID
The `main_lineage` field contains "lineage"	DNA of Mycobacterium tuberculosis species detected
The `main_lineage` or `sub_lineage` field contains "BCG"	DNA of Mycobacterium bovis BCG detected
The `main_lineage` or `sub_lineage` field does **not** contain "BCG" but ***does*** contain "bovis" or "La1"	DNA of Mycobacterium bovis (not BCG) detected
The `main_lineage` field is blank, "NA", or does not exist	DNA of Mycobacterium tuberculosis complex detected

5.4.2. ELSE "DNA of Mycobacterium tuberculosis complex NOT detected"

5.5. **<u>MTBC ID by tNGS only:</u>** Parse Coverage and Laboratorian reports to populate LIMS report field "Lineage ID" for species ID:

5.5.1. IF ≥ min_percent_loci_covered of genes on the Coverage report that are ALSO on the LIMS report have ≥ min_percent_coverage (per percent_coverage field), **THEN** assign Lineage ID bz`ased on the following table:

Information coming from tbp-parser Laboratorian report	**In LIMS output of tbp-parser**
	Lineage ID
pncA contains a "p.His57Asp" mutation that passes positional QC	DNA of Mycobacterium bovis BCG detected
pncA contains a "p.His57Asp" mutation that **does NOT** pass positional QC	DNA of Mycobacterium tuberculosis complex detected (M. bovis not ruled out)
pncA **does NOT** have a "p.His57Asp" mutation	DNA of Mycobacterium tuberculosis complex detected (not M. bovis)

5.5.2. ELSE: "DNA of Mycobacterium tuberculosis complex NOT detected"

6. <u>Looker output</u>

6.1.1. All gene-drug combinations are used; none are excluded.

6.1.2. "ID" field: copy over the value that is generated by TBP Parser for "Lineage ID" in the LIMS report (see section 5.4).

6.1.3. "lineage" field: populate using `main_lineage` output of TB Profiler (see section 5.2.2 above).

7. Additional tNGS-specific logic

7.1. **IF** a mutation appears **outside** the gene locus ranges provided by the `--coverage_bed` input file:
Add "This mutation is outside the expected region" to the warning field
AND assign
rationale = "NA"
confidence = "NA"
looker_interpretation = "NA"
mdl_interpretation = "NA"
AND ignore this mutation in the LIMS report entirely

7.2. Breadth of coverage QC for non-overlapping primers uses **the aggregated result** of the segments and fails the mutation if the **aggregate** does not meet QC thresholds.

Table 1: Proximal promoter regions defined by the WHO v2 catalogue; the regions for mmpR5 (Rv0678) were designated by CDPH.

	Promoter region coordinates
aftB	-129 to -1
ahpC	-93 to -1
atpE	-51 to -1
bacA	-81 to -1
ccsA	-191 to -1
clpC1	-106 to -1
ddn	-51 to -1
dnaA	-314 to -1
eis	-84 to -1
embA	-86 to -1
embC	-1982 to -1
embR	-103 to -1
ethA	-51 to -1
ethR	-26 to -1
fbiA	-138 to -1
fbiC	-127 to -1
fgd1	-51 to -1
gid	-79 to -1
glpK	-52 to -1
gyrA	-35 to -1
gyrB	-108 to -1
hadA	-51 to -1
inhA	-813 to -1
katG	-532 to -1
*mmpR5 (Rv0678)**	-84 to -1
mmpS5	-85 to -1
mshA	-669 to -1
mtrA	-376 to -1
mtrB	-50 to -1
ndh	-96 to -1
nusG	-201 to -1
panD	-51 to -1 **and** -1949 to -1838
pepQ	-51 to -1
pncA	-51 to -1
PPE35	-122 to -1
rplC	-51 to -1 **and** -503 to -323
rpoA	-536 to -1
rpoB	-263 to -1
rpoC	-45 to -1
rpsA	-100 to -1
rpsL	-234 to -1
rrl	-51 to -1
rrs	-151 to -1
Rv0010c	-156 to -1
Rv0565c	-78 to -1

Rv1129c	-51 to -1
Rv1258c	-58 to -1
Rv1979c	-470 to -1
Rv2477c	-88 to -1
Rv2680	-153 to -1
Rv2681	-2 to -1
Rv2752c	-51 to -1 **and** -984 to -934
Rv2983	-51 to -1

Rv3083	-51 to -1
Rv3236c	-51 to -1 **and** -538 to -488
sigE	-51 to -1
tlyA	-51 to -1 **and** -236 to -185
tsnR	-51 to -1
ubiA	-51 to -1
whiB6	-126 to -1
whiB7	-404 to -1